

Exp no 5 : Calculation of Group velocity, Phase velocity and phase constant of EM waves in rectangular waveguides

Test case 1 :

clc;

clear;

close;

// TE10 Mode

a = 8.636*10^-2;

b = 4.318*10^-2;

f = 8*10^9;

V = 3*10^8;

m = input("Enter the value of m = ");

n = input("Enter the value of n = ");

// Cutoff frequency

fc = (V/(2*pi))*sqrt((m*pi/a)^2 + (n*pi/b)^2);

disp("Cutoff Frequency (GHz) = ", fc*10^-9);

// Phase velocity

A = V/sqrt(1-(fc/f)^2);

disp("Phase Velocity (Mm/sec) = ", A*10^-6);

// Group velocity

B = V^2/A;

disp("Group Velocity (Mm/sec) = ", B*10^-6);

Scilab 2025.1.0 Console

File Edit Control Applications ?

Scilab 2025.1.0 Console

enter the value of m= 1

enter the value of n= 0

"Cut off frquency="

1.7369152

"Phase velocity in Mm/sec="

333.03634

"Group velocity in Mm/sec ="

270.24078

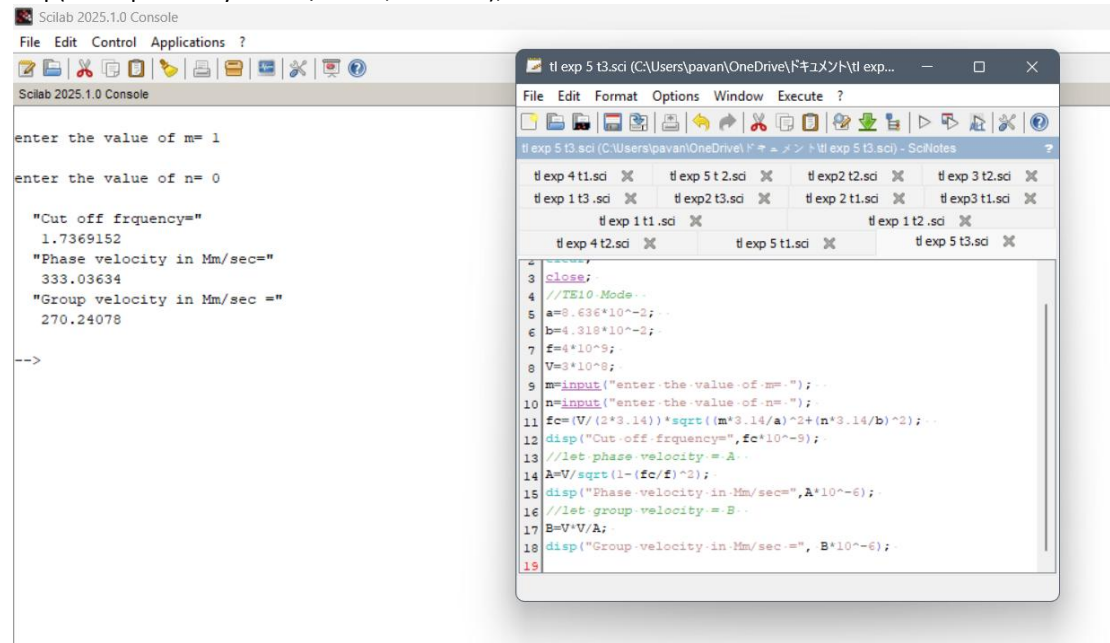
-->

The screenshot shows the Scilab 2025.1.0 interface. The console window on the left displays the output of the script for m=1 and n=0, showing the cutoff frequency, phase velocity, and group velocity. The editor window on the right shows the source code of the script, which includes comments for TE10 mode and calculations for cutoff frequency, phase velocity, and group velocity.

```
1 close;
2 //TE10 Mode
3 a=8.636*10^-2;
4 b=4.318*10^-2;
5 f=8*10^9;
6 V=3*10^8;
7 m=input("enter the value of m=");
8 n=input("enter the value of n=");
9 fc=(V/(2*pi))*sqrt((m*pi/a)^2+(n*pi/b)^2);
10 disp("Cut-off frequency=",fc*10^-9);
11 //let phase velocity = A
12 A=V/sqrt(1-(fc/f)^2);
13 disp("Phase velocity in Mm/sec=",A*10^-6);
14 //let group velocity = B
15 B=V^2/A;
16 disp("Group velocity in Mm/sec =", B*10^-6);
```

Test case 2:

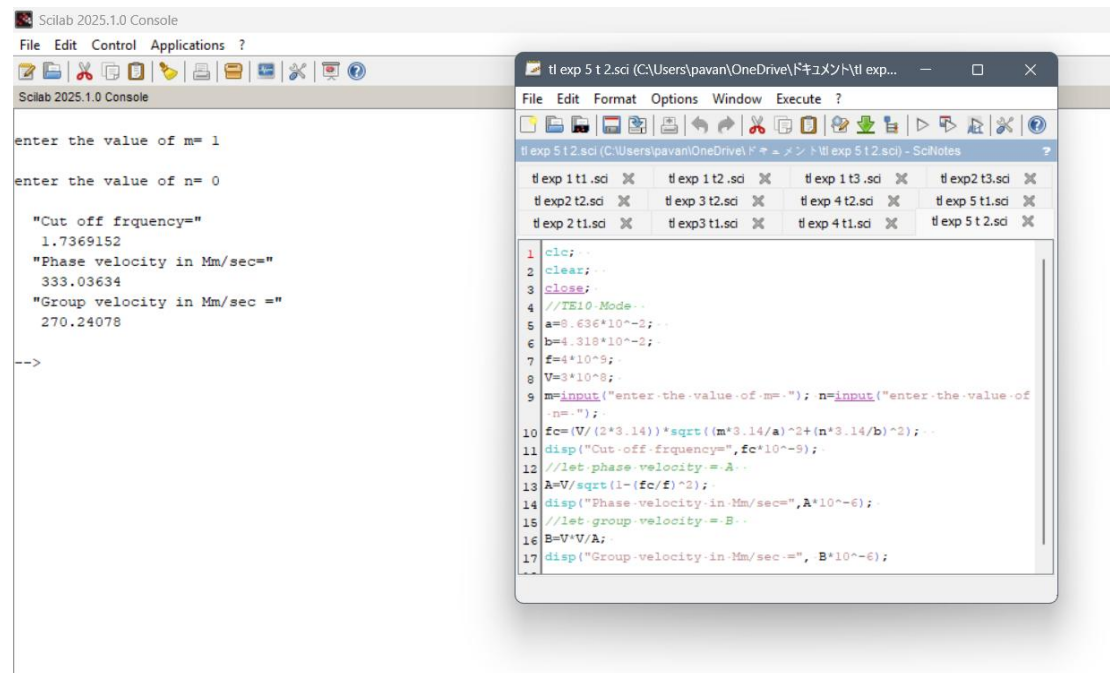
```
clc;
clear;
close;
//TE10 Mode
a=8.636*10^-2;
b=4.318*10^-2;
f=4*10^9;
V=3*10^8;
m=input("enter the value of m= "); n=input("enter the value of n= ");
fc=(V/(2*3.14))*sqrt((m*3.14/a)^2+(n*3.14/b)^2);
disp("Cut off frequency=",fc*10^-9);
//let phase velocity = A
A=V/sqrt(1-(fc/f)^2);
disp("Phase velocity in Mm/sec=",A*10^-6);
//let group velocity = B
B=V*A/A;
disp("Group velocity in Mm/sec =", B*10^-6);
```



Test case 3 :

```
clc;
clear;
close;
//TE10 Mode
a=8.636*10^-2;
b=4.318*10^-2;
f=4*10^9;
V=3*10^8;
m=input("enter the value of m= ");
n=input("enter the value of n= ");
fc=(V/(2*3.14))*sqrt((m*3.14/a)^2+(n*3.14/b)^2);
disp("Cut off frequency=",fc*10^-9);
//let phase velocity = A
A=V/sqrt(1-(fc/f)^2);
disp("Phase velocity in Mm/sec=",A*10^-6);
//let group velocity = B
B=V*A/A;
```

```
disp("Group velocity in Mm/sec =", B*10^-6);
```



Test case 4:

```
clc;
clear;
close;
//TE10 Mode
a=8.636*10^-2;
b=4.318*10^-2;
f=4*10^9;
V=3*10^8;
m=input("enter the value of m=");
n=input("enter the value of n=");
fc=(V/(2*3.14))*sqrt((m*3.14/a)^2+(n*3.14/b)^2);
disp("Cut off frequency=",fc*10^-9);
//let phase velocity = A
A=V/sqrt(1-(fc/f)^2);
disp("Phase velocity in Mm/sec=",A*10^-6);
//let group velocity = B
B=V*A/f;
disp("Group velocity in Mm/sec =", B*10^-6);
```

